

Tunis, Tunisia

+216 22 296 652

maramnuiguess@gmail.com

linkedin.com/in/maram-nouiguez-ab929a294

Key Skills:

Machine Learning • Deep Learning • Computer Vision • Python (OpenCV, TensorFlow, scikit-learn, Pandas) • Big Data • SQL • Data Visualization

Professional Profile

Final-year (5th year) Industrial Computing and Automation student at INSAT, specializing in Intelligent Systems (Machine Learning, Deep Learning, Big Data, Data Science).

I like applying AI where it actually saves someone time on the floor: at Autoliv, part of the planning tool I built for the VS Matrix method used an AI-assisted compatibility matrix, one of the pieces that helped cut a 4-day manual task down to under 20 minutes.

Looking for a final-year internship in AI applied to industry or industrial data science, starting February 2027.

Education

Engineering Diploma, Industrial Computing and Automation – Intelligent Systems track

2022 – 2027

INSAT – National Institute of Applied Sciences and Technology, Tunis (Machine Learning, Big Data, Deep Learning, Data Science, Operations Research, UML)

Professional Experience

Intern, APS Department (Autoliv Production System – Continuous Improvement & Digitalization)

Jun – Aug 2025

Autoliv, El Fahs, Tunisia – supervised by Slim Hsan, APS Manager

- Built a **Python/Streamlit** planning app for the VS Matrix method that included an **AI-assisted compatibility matrix** (AG Grid, session state), contributing to cutting a 4+ working-day, cross-department task down to **under 20 minutes for one person**.
- Worked with real production data (Excel-extracted) to feed the tool's KPI dashboards (OEE, PPH) – hands-on experience turning raw plant data into something a model or dashboard can use.

Intern, Industrial Engineering

2024

Lacroix Electronics, Tunisia

- Exposure to PLC/sensor data streams (Siemens TIA Portal) while helping diagnose a robot arm fault – a concrete look at where sensor data quality affects automated decision-making.

Technical Projects

VoltMind – AI Assistant for Electrical Circuit Design – built an intelligent tool to help students design and simulate electrical circuits (university project).

Embedded Industrial Vision System for Quality Control – Raspberry Pi 5, OpenCV, Python; automatic visual sorting, sub-millimetric label-position checking, and OCR-based manufacturing-date validation.

Autonomous Industrial Inspection Robot – ROS2 Humble, Gazebo, Nav2, ESP32/micro-ROS, PyQt6, Vosk offline speech recognition; SLAM-based autonomous mapping and navigation, supervised by Pr. Saber Tlili.

Industrial Digital Twin – SolidWorks, Simulink/MATLAB, LabVIEW/Arduino; three-layer synchronized digital twin (3D model, dynamic flow simulation, real-time control).

Additional projects: SAP production planning, AnyLogic simulation, Factory IO/TIA Portal automation, operations research logistics study, project management project, Spark (educational STEM app).

Technical Skills

AI & Machine Learning:

Machine Learning, Deep Learning, Big Data, Data Science, Computer Vision

Programming & Data:

Python (OpenCV, TensorFlow, scikit-learn, Pandas), SQL, MATLAB

Robotics & Simulation:

ROS2, Gazebo, Nav2, Simulink, LabVIEW

Reporting:

Power BI, Excel (VBA)

Languages:

Arabic (native), French (C1), English (C1)

Leadership & Distinctions

Chairwoman, IEEE Women in Engineering – INSAT SAG (2024–2025) – led a **100+ member** organization: budget, volunteer team, and end-to-end technical events.

Team Lead, VIP & Protocol Department – National Robotics Week 8.0 (2026) – ran the VIP & Protocol team and developed panel content (automotive/SDV, cyber-physical security) for a national robotics event.

Robotics Instructor, Play Mind Club (2025 – Jun 2026) – ran STEM workshops (Scratch, Arduino, robotics) for children aged 8–12.